ELEC3705 Fundamentals of Quantum Engineering Laboratory 6 : Quantum Tunnelling and Evanescent Waves

Abstract

In this lab you will observe quantum tunnelling and evanescent waves through numerical simulations. You will learn to understand differential equations in matrix form and use frequency domain techniques (i.e. constructing a wavepacket in reciprocal space) to define a particle's initial state.

Introduction

Today we will revisit the time-independent Schrödinger equation for an electron encountering a potential barrier in 1-dimensional 'free' space.

Matrix formulation of a PDE Hamiltonian

The time-independent Schrödinger equation for an electron in a Coulomb potential is given by

$$\underbrace{\left[\frac{-\hbar^2}{2m_0}\nabla^2 + V(x)\right]}_{H}\Psi(x) = E\Psi(x) \tag{1}$$

where $\nabla^2 = \dfrac{\partial^2}{\partial x^2}$ is the Laplacian operator, m_0 is the particle's rest mass, and V describes the Coulomb potential our particle is moving through. Here we have applied the same separation of variables seen in lectures. You should recognize that here $\Psi(x)$ is a stationary state of the system Hamiltonian, i.e. an eigenstate of the system, with associated energy E. In this lab we will learn to translate a linear partial differential equation into the more computer-friendly matrix mechanics we have used so far.

We begin by considering that our 1-dimensional space is discretised into N regions, each with width dx. We consider the basis states to be each of these N regions, so the spatial basis functions look like single impulses with width dx at each discrete position in the range $(0, N \cdot dx)$. At each position x, the contribution of the Coulomb potential to the total energy

of our particle is simply $V(x)$. This is an energy detuning, which are expressed in our $N \times N$ Hamiltonian matrix along the main diagonal, as in previous labs.

$$V(x) \Rightarrow \begin{pmatrix} V(0) & 0 & 0 & \cdots & 0 \\ 0 & V(dx) & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & V(N \cdot dx) \end{pmatrix} \tag{2}$$

We now consider the other component of our system Hamiltonian, $\frac{-\hbar^2}{2m_0}\nabla^2$. Here $\frac{-\hbar^2}{2m_0}$ is simply a constant, and so we can just consider the representation of $\nabla^2 = \frac{\partial^2}{\partial x^2}$ as a matrix. Since we are working in 1 dimension, the second partial derivative can be obtained from first principles as

$$\frac{\partial^2}{\partial x^2} f(x) = \lim_{\partial x \to 0} \frac{f(x + dx) - 2f(x) + f(x - dx)}{dx^2} \tag{3}$$

When applying this numerically, the smallest ∂x we can use is exactly dx. Therefore, what the Laplacian operator at a point x really does is take a linear combination of the wavefunction at point x and the points immediately to the left and right of x. Therefore, the Laplacian operator in our matrix formalism will have non-zero components only in the main diagonal and the first off-diagonals. Note that we forgo a periodic boundary condition, so the first and last well actually 'leak' energy out of the system.

$$\nabla^2 \Rightarrow \frac{1}{dx^2} \begin{pmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 & -2 \end{pmatrix} \tag{4}$$

Finally, we have our system Hamiltonian fully described

$$H = \frac{-\hbar^2}{2m_0}\frac{\partial^2}{\partial x^2} + V(x) \Rightarrow \begin{pmatrix} V(0) - 2L & L & 0 & 0 & \cdots & 0 \\ L & V(1) - 2L & L & 0 & \cdots & 0 \\ 0 & L & V(2) - 2L & L & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & L & V(N \cdot dx) - 2L \end{pmatrix} \tag{5}$$

where $L = \frac{-\hbar^2}{2m_0 \cdot dx^2}$.

Defining a moving particle in 1-dimension

It is quite trivial to define a wavepacket in 1-dimensional space as a Gaussian distribution about some point x_0 with a standard deviation of σ.

$$\psi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2}\right) \tag{6}$$

However, how do we now give this wavepacket an initial velocity? This is a more complicated matter that can be resolved by thinking in terms of spatial *frequency*, k, rather than in x. As you know, the momentum of a particle is given by $p = \hbar k = mv$, so there is a natural relationship between the momentum of a particle and its location in k-space. We define our k-space from the discrete Fourier transform of $\psi(x)$, $\phi(k) = \mathcal{F}[\psi(x)]$. It is worth noting that the Fourier transform of a Gaussian distribution is another Gaussian distribution in the mapped space, so if we construct a Gaussian in k-space and then transform back to real-space, we will fortunately still have a Gaussian wavepacket.

Question 1

Show that the Fourier transform of a Gaussian, g(x) = e^{-ax^2} is another Gaussian.

Question 2

Suppose our x-axis ranges from 0 to 1 µm in steps of 1 nm, what will the resolution and range of k-space be? Further, if we let $\psi(x)$ be described by a Gaussian with $\sigma = 50\,\text{nm}$, what will σ_k, i.e. the standard deviation in k-space, be?

Natural Units

For the remainder of this lab, we will be making use of a convenient system for our units known as the 'natural units'. For example, we will set the reduced Planck constant $\hbar = 1$. Since we were always cancelling out $\hbar$ when doing time-evolution, we may as well eliminate it altogether. An added benefit is that we reduce the risk of having floating-point precision errors, since we have the best resolution near to 0. We will also do this for all of our dimensions, minimising the exponents of each of our terms.

For the Laplacian term of our Hamiltonian, in real units, we would have a barrier on the order of 1 nm in length, $\hbar$ would be $\approx 1\times10^{-31}$ J s, and the mass of an electron is $\approx 1\times10^{-31}$ kg. This yields $\dfrac{1}{dx^2}\dfrac{-\hbar^2}{2m_0} \approx 1 \times 10^{-19}$ J which is approximately an electron volt. We can simplify our calculations by replacing all of these quantities with 1 in their respective units, and doing the same for the Coulomb potential $V(x)$.

This allows us to then simplify some of our operators:

- $E = \frac{\hbar^2 k^2}{2m_0} \Rightarrow \frac{k^2}{2}$
- $\frac{-\hbar^2}{2m_0}\nabla^2 \Rightarrow \frac{1}{2}\frac{\partial^2}{\partial x^2}$

Tunnelling and Transmission

The quantum phenomenon of tunnelling can best be quantified by measuring the probability that an incident particle will be found on the other side of a potential barrier after some time T. For a particle in free space, we can let $T \Rightarrow \infty$ since there will only be a single collision with the barrier, however for a bounded particle we have to be more careful because a particle may collide with the barrier multiple times.

Question 3

How do you measure the probability of your particle being transmitted through the barrier?

Task

We will consider an electron in a bounded potential well (surrounded by infinite energy barriers), with a single finite energy barrier inside of width 10 m and height 1 J. Create a Hamiltonian given the dimensions shown in Figure 1, dividing the x-axis into $N = 400$ parts. You will then need to create a wavepacket in k-space for the electron. Let $\phi(k)$ follow a Guassian with centre $k_0 = 2\,\mathrm{rad\,m^{-1}}$ and variance $\sigma_k = 0.2\,\mathrm{rad\,m^{-1}}$.

Calculate the energy of the particle using $\phi(k)$.

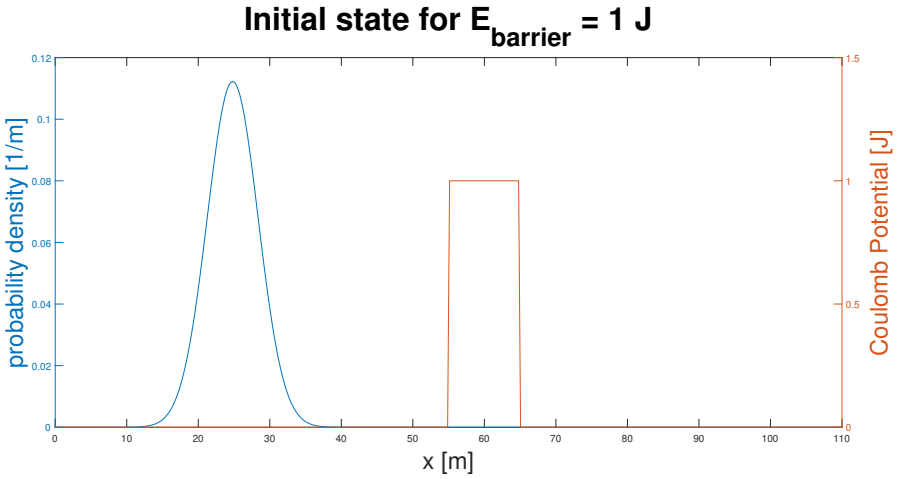


Figure 1: The setup of the single electron system.

Use the inverse Fourier transform to reconstruct $\psi(x) = \mathcal{F}^{-1}[\phi(k)]$. Note that the Fourier transform is translation invariant (ignoring phase), so we can translate $\psi(x)$ without losing the meaning of $\phi(k)$. Set the centre of your wavepacket in space to be $x_0 = 25\,\mathrm{m}$ Perform a time-evolution of your spatial wavefunction for $T = 30\,\mathrm{s}$ with $dt = 0.5\,\mathrm{s}$. Use a differential unitary time evolution operator $U(dt) = \exp\left(-i\frac{H}{\hbar}dt\right)$ to incrementally evolve the wavepacket.

Once you have the basics of this lab task working, you should evaluate the transmission of your particle as a function of the barrier height. Vary the barrier height from 0 to 3 J and plot the probability that the particle will be transmitted.

Some Python functions you will be making use of in this lab are

- `numpy.heaviside(x - x_0)` is the unit-step function about `x_0`.
- `numpy.fft.ifft(phi)` will give you the inverse fast Fourier transform of your $\phi(k)$, which you will define.
- `numpy.fft.fftshift(phi)` swap half space of the spectrum and center it around the mid point.

- `numpy.diag(vec, k)` will create a matrix with the vector `vec` on the `k`th diagonal.
- `numpy.roll(phi, N)` will roll an array by N elements towards the right. Can be useful in order to translate phi in both x or k space.

Extended Task

Replace your single potential barrier with two distinct energy barriers of the same height and each with half of the width, separated by the width. Now instead of varying the barrier height, you should vary k_0, and hence plot transmission as a function of the electron's kinetic energy.